

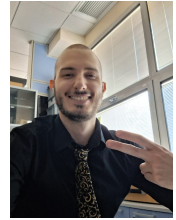
# Coser Omar

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in Omar Coser

omarcoser



## Work Experiences

April 2025 – 31 December 2026

### ■ Guest Contract Max Planck Institute for Intelligent Systems – Personal Projects

Since the end of my PhD contract I decided to start working on my idea given the possibility to use the Computational Resources of the Max Planck Institute for Intelligent Systems to implement them.

Project 1: **ELISA: An Interpretable Hybrid Agent for Expression-Grounded Discovery in Single-Cell Genomics** (Accepted on Briefing in Bioinformatics, IF:7.3)

Project 2: **Cells as distributions: local covariance improves graph-based clustering but not transition detection** (Submitted at the NeurIPS 2026 I Can't Believe It's Not Better: Failure Modes of AI in Biology Workshop)

Project 3: **Pre-training Barely Beats Random Initialisation for Single-Cell Disease Classification** (Submitted at the NeurIPS 2026 I Can't Believe It's Not Better: Failure Modes of AI in Biology Workshop)

Project 4: **Working on a New Model for Time Series Classification**

Oct 2024 – Sept 2026

### ■ Visiting PhD Student in the Deep Models and Optimization, (Max Planck Institute for Intelligent Systems)

During this period I Am working on the transformer architecture and in particular I am working on the self-pre-training of it

Project Title: **Towards Understanding Self-Pretraining for Sequence Classification** (Submitted at NeurIPS 2026 Available on Arxiv.. Received a score of 4, 4, 5 (Where 4 is borderline accept and 5 is accept from three reviewer before rebuttal, notification 23 September)

Project 2: **Introducing Self-Pretraining useful to improve diagnosis in medical Time Series**, (Submitted at NeurIPS Workshop Responsible Communication of Machine Learning Research in Biomedicine, available on Arxiv)

## Work Experiences (continued)

- Jan 2023 – June 2026  **Unit of Advanced Robotics and Human-Centered Technologies, Unit of Artificial Intelligence and Computer Systems (Università Campus Bio-medico di Roma)**  
Project 1 (Ended): **AI-based methodologies for exoskeleton-assisted rehabilitation of the lower limb: A review**(Literature Review)  
Project 2: **Deep Learning for Human Locomotion Analysis in Lower-Limb Exoskeletons: A Comparative Study** (Published on Frontiers in Computer Science)  
Project 3: **Introducing Self-Prerraining usefull to improve diagnosis in medical Time Series**, (Submitted at NeurIPS Workshop Responsible Communication of Machine Learning Research in Biomedicine, available on Arxiv)
- Jan 2022 – Nov 2022  **Structured Machine Learning Lab (Università di Trento)**  
Development of a web application empowered with Reinforcement Learning. Reinforcement learning part allow to elicit user recommended tutoring session (final outcome is an Intelligent Tutor). During this period I used Python (flask), Javascript, html, css as programming language. Moreover the application was connected to a small database in MySQL
- Feb 2021 – Sep 2021  **Laboratory of Translational Architectomics (FBK)**  
During this period I used Machine Learning techniques (k-means, hierarchical clustering, analyzed of graph data (Problem solved: Sub-graph isomorphism problem, Travelling Sales-man Problem). To study the geometry of polysomes structure.
- June 2019 – Sep 2019  **Bianconi Lab, CIBIO**  
Main Topic: Multi-drug resistance Bacteria. Worked with Bacteria. I have performed different biotechnological analysis like MIC assay, Western Blotting, DNA/RNA and protein extraction

## Education

- Jan 2023 – Jan 2026  **National PhD In Artificial intelligence** Università Campus Biomedico Rome. Currently in the Unit of Artificial Intelligence and Computer Systems and the Unit of Advanced Robotics and Human-Centered Technologies  
**Exams Attended (total of 140 hours):** "Mathematics, deep learning, deep reinforcement learning", "Probabilistic Graphical Models in Intelligent Systems","Can we trust AI? Opening the blackbox", "Behavioural biometrics for healthcare, security and related field", "Hand on Machine and Deep Learning", "Human-Centered AI (HCAI): an innovative vision for designing intelligent systems"
- 2019 – 2021  **Master Degree in Quantitative and Computational Biology, University of Trento, 110/110**  
**Relevant Exams:** Scientific Programming, Biostatistics, Biological Networks and Data Analysis, Data Mining (Machine Learning and Biological Data Mining), Bioinformatics, Biotechnonology Engineering, Mathematical Modelling, Microbial and Human Genomics.

## Education (continued)

2016 – 2019

📖 **Bachelor in Biotechnology, University of Trento**

**Relevant Exams:** Microbiology, Cell Biology, Molecular Biology, Cellular and Microbial Biotechnology, Physiology, Organic and Inorganic Chemistry, Computer Science, Computational Biology.

## Research Publications

### Journal and Conference Articles

- 1 O. Coser, C. TAMANTINI, P. Soda, and L. ZOLLO, “Ai-based methodologies for exoskeleton-assisted rehabilitation of the lower limb: A review,” *Frontiers in Robotics and AI*, vol. 11, p. 1 341 580,
- 2 O. Coser, “Artificial intelligence applications to lower limb rehabilitation robotics,” 2026.
- 3 O. Coser, “Elisa: An interpretable hybrid generative ai agent for expression-grounded discovery in single-cell genomics,” *Accepted in Briefing in Bioinformatics waiting to get public, arXiv preprint arXiv:2603.11872*, 2026.
- 4 O. Coser, A. Orvieto, P. Soda, and L. Zollo, “Is self-pretraining really useful to improve diagnosis in medical time series?” *arXiv preprint arXiv:2608.06122*, 2026.
- 5 O. Coser, L. Zollo, P. Soda, and A. Orvieto, “Towards understanding self-pretraining for sequence classification,” 2026. arXiv: 2605.21070 [cs.LG]. 🔗 URL: <https://arxiv.org/abs/2605.21070>.
- 6 O. Coser and A. Orvieto, “Towards understanding self-pretraining for sequence classification,” *ES-FoMo III: 3rd Workshop on Efficient Systems for Foundation Models- International Conference on Machine Learning*, 2025.
- 7 O. Coser, C. Tamantini, M. Tortora, *et al.*, “Deep learning for human locomotion analysis in lower-limb exoskeletons: A comparative study,” *Frontiers in Computer Science*, vol. Volume 7 - 2025, 2025. 🔗 DOI: 10.3389/fcomp.2025.1597143.
- 8 F. Aksu, A. Bria, A. N. Caragliano, *et al.*, “Towards ai-driven next generation personalized healthcare and well-being,” vol. 3762, pp. 360–365, 2024.

## Skills

|                                        |                                                                                                                                                        |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages                              | 📖 Strong reading, writing and speaking competencies for English and Italian.                                                                           |
| Coding                                 | 📖 Python (Libraries: Pytorch, Tensorflow, sklearn, numpy, pandas, matplotlib), Basic of R, $\text{\LaTeX}$ .                                           |
| Artificial Intelligence                | 📖 AI Agents, LLM, Neural Network Models, Transformer model, Pretraining methods, Multimodal Fusion, Condor and Slurm based High Performance Computing. |
| Quantitative and Computational Biology | 📖 Basic Knowledge about Omics and bioinformatics analysis                                                                                              |
| Data                                   | 📖 Time Series, Multimodal-Multivariate Time Series, Graph Data, Tabular Data.                                                                          |

## Miscellaneous Experience and Award

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### Conferences

- 28 June- 1 July 2021     Participation in the XXV Congress of the SIBPA, Poster session, Poster Title: Ultrastructural analysis of ribosome organization in large polysome
- 4 December 2025     Participation at the MedicinAI Workshop organized by UniTN during which I presented a work entitled "Is Self-PreTraining really useful in medical time series? A transformer based study".
- 2026     ICML 2026 Reviewer.
-  ICLR 2026 Workshop on Generative AI for Genomics Reviewer.
-  ICML 2026 Workshop on Generative and Agentic AI for Biology Reviewer.

### Awards

- March 2025     My paper "AI-based methodologies for exoskeleton-assisted rehabilitation of the lower limb: A review", has been selected among the 11 outstanding papers published in Frontiers in Robotics and AI in 2024 out of 261 papers published.
- July 2026     Briefings in Bioinformatics (Oxford University Press) granted a full waiver of the open access publication charge (>€4,000) for my paper "ELISA: An Interpretable Hybrid Generative AI Agent for Expression-Grounded Discovery in Single-Cell Genomics".

## References

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### Prof. Antonio Orvieto

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### Prof. Paolo Soda

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### Omar Coser

Trento, September 9, 2026